

8.2 Diffraction & Interference

Question Paper

Course	CIE A Level Physics
Section	8. Superposition
Topic	8.2 Diffraction & Interference
Difficulty	Hard

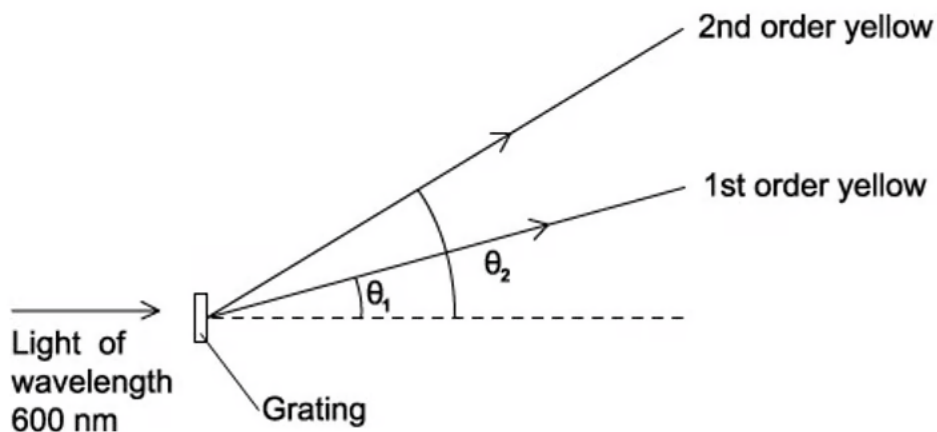
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Yellow light with a wavelength of 600 nm is used in a diffraction grating experiment. The grating has a separation of $2.00 \mu\text{m}$.



What would be the angular separation between the first order and second order maxima?

- A. 54.3°
- B. 36.9°
- C. 19.4°
- D. 17.5°

[1 mark]

Question 2

Yellow light with a wavelength of 600 nm is used in an experiment to show interference fringes produced by double-slit interference. The fringe separation is 4.0 mm and the separation of the slits is 0.60 mm.

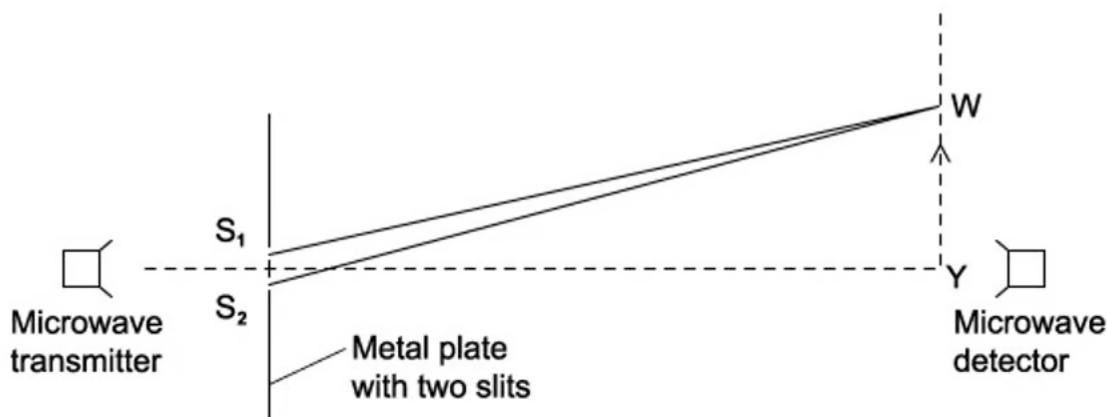
What would be the distance between the double slit and the screen?

- A. 4.0 m
- B. 2.5 m
- C. 0.4 m
- D. 0.25 m

[1 mark]

Question 3

An experiment was set up to investigate two-source interference. Microwaves of wavelength λ passed through S_1 and S_2 .



The detector is moved in the direction of the arrow from Y to W. The signal detected decreased to point W then starts to increase again as it continues past W.

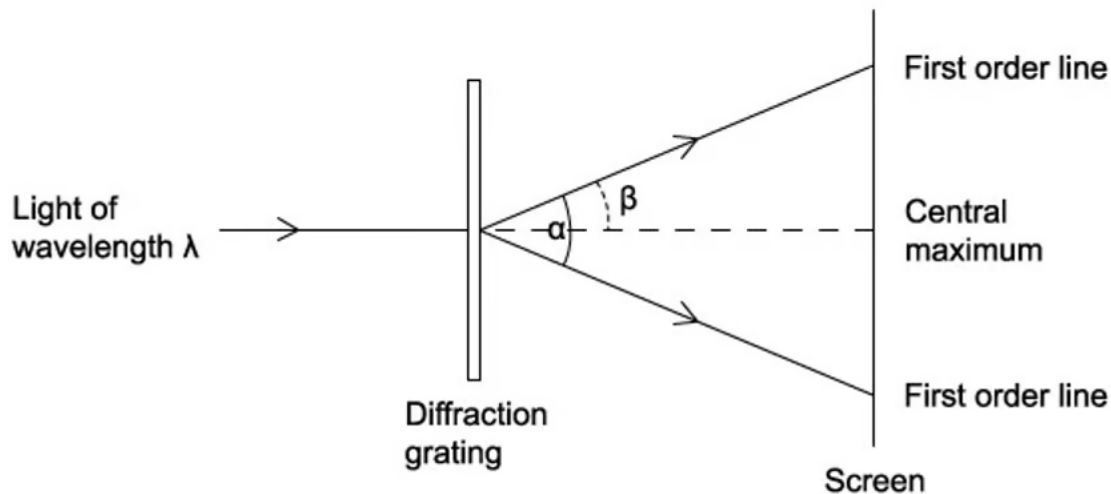
Which of the following would be the equation to determine the position of point W?

- A. $S_2 W - S_1 W = \frac{\lambda}{2}$
- B. $S_2 W - S_1 W = \lambda$
- C. $YW = \frac{\lambda}{2}$
- D. $YW = \lambda$

[1 mark]

Question 4

An experiment was set up to investigate light passing through a diffraction grating with slit spacing d . The fringe pattern was observed on a screen.



Which of the following gives the angle between the two first-order lines?

A. $2 \sin^{-1}\left(\frac{\lambda}{2d}\right)$

B. $2 \sin^{-1}\left(\frac{\lambda}{d}\right)$

C. $\sin^{-1}\left(\frac{\lambda}{2d}\right)$

D. $\sin^{-1}\left(\frac{\lambda}{d}\right)$

[1 mark]

Question 5

Yellow light with a wavelength of 600 nm is used in an experiment to show interference fringes produced by double-slit interference. The fringe separation is 4.0 mm on the screen.

What would be the fringe spacing be if the wavelength is changed to 400 nm and the separation of the slits is doubled?

- A. 12.0 mm
- B. 5.3 mm
- C. 3.0 mm
- D. 1.3 mm

[1 mark]**Question 6**

A student carried out an experiment on monochromatic light with a wavelength of 5.30×10^{-7} m. They set up the source so that it was incident normally to the diffraction grating.

The student observed that the first-order maximum of was 15.4° to the direction of the incident light.

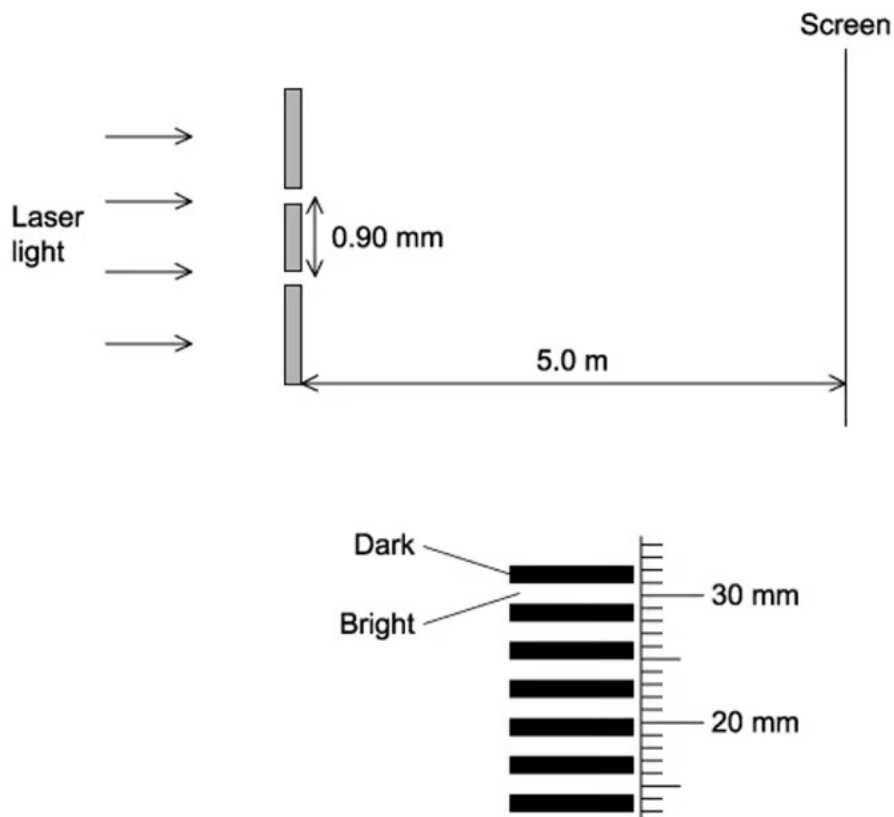
What would be the angle between the first order and second order maxima?

- A. 32.1°
- B. 16.7°
- C. 15.4°
- D. 7.6°

[1 mark]

Question 7

A student carried out an experiment on Young's slits using a laser light. There is also a section of the pattern produced with a ruler placed next to it.



What is the wavelength of the light?

- A. 540 nm
- B. $3.4 \mu\text{m}$
- C. 480 nm
- D. $3.2 \mu\text{m}$

[1 mark]

Question 8

A student carried out an experiment on interference patterns from a diffraction grating and double-slit and compared the results.

Using the diffraction grating, yellow light with the first order at 30° to the normal and the grating was seen.

Using the same light interference fringes were seen on a screen 1.0 m from the double slit. The slit separation was 500 times greater than the line spacing of the grating.

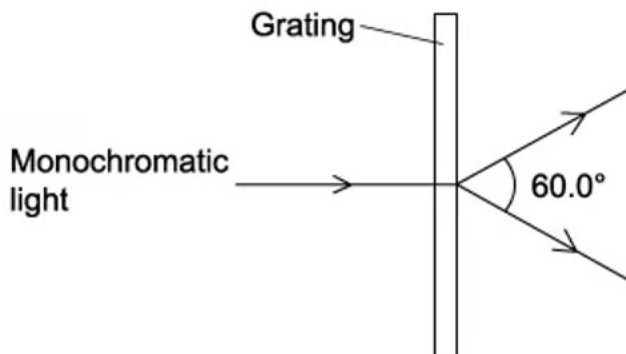
What fringe separation did the student see on the screen?

- A. 10 cm
- B. 1 mm
- C. 250 nm
- D. $10\ \mu\text{m}$

[1 mark]

Question 9

The wavelength of monochromatic light can be determined using a diffraction grating. This is shown in the diagram.



The angle between the first-order diffraction maxima is 60° , and the spacing of the slits is $1.15 \times 10^{-6}\text{ m}$.

What is the wavelength of the light?

- A. 575 nm
- B. 498 nm
- C. 996 nm
- D. 287 nm

[1 mark]

Question 10

When a parallel beam of white light is incident normally on a diffraction grating the second-order and third-order spectra partially overlap.

When the second order wavelength is 600 nm, which of the following third-order wavelengths would appear at the same angle?

- A. 900 nm
- B. 600 nm
- C. 400 nm
- D. 300 nm

[1 mark]

Model Answers: Hard

1

The correct answer is **C** because:

- The diffraction grating equation is given by
 - $d \sin \theta = n\lambda$
- For $n=1$
 - $d \sin \theta_1 = n\lambda$
 - $\sin \theta_1 = \frac{\lambda}{d} = \frac{600 \times 10^{-9}}{2.00 \times 10^{-6}} = 0.30$
 - So, $\theta_1 = \sin^{-1} 0.30 = 17.45^\circ$
- For $n=2$
 - $d \sin \theta_2 = 2\lambda$
 - $\sin \theta_2 = \frac{2\lambda}{d} = 0.60$
 - So, $\theta_2 = \sin^{-1} 0.60 = 36.87^\circ$
- Angular separation is equal to $(\theta_2 - \theta_1)$, so:
 - $(\theta_2 - \theta_1) = 36.87^\circ - 17.45^\circ = 19.42^\circ$
- Therefore, the angular separation is equal to 19.4°

2

The correct answer is **A** because:

- The fringe separation equation is given by
 - $x = \frac{\lambda D}{a}$
- Where:
 - x is the fringe separation = 4.0 mm
 - a is the slit separation = 0.60 mm
 - λ is the wavelength = 600 nm
- So, rearranging for distance D :

$$\circ D = \frac{xa}{\lambda} = \frac{(4.0 \times 10^{-3})(0.60 \times 10^{-3})}{600 \times 10^{-9}} = 4.0 \text{ m}$$

- Therefore, the distance between the double slit and the screen is 4.0 m

3

The correct answer is **A** because:

- A minima is found at W since the signal keeps on decreasing until it reaches W, so, at W, the signal has the lowest amplitude
 - Therefore, W is the first minima: this is where destructive interference takes place
- S_2W and S_1W are the distances between the slits and the screen, hence the path difference is equal to $S_2W - S_1W$
- The condition for destructive interference is path difference $\frac{n\lambda}{2}$
- For the first minima ($n = 1$):
 - Path difference $= \frac{\lambda}{2}$
 - $S_2W - S_1W = \frac{\lambda}{2}$
- So, the distance $\frac{\lambda}{2}$ is equal to the path difference between S_2W and S_1W which is $S_2W - S_1W$

B is incorrect because this would be at the first maxima, where path difference $= n\lambda$

C & D are incorrect because

Fringe separation is equal to $x = \frac{\lambda D}{a}$

The distance between the central maxima (Y) and the first minima (at W) is half the fringe separation $YW = \frac{x}{2}$

So, YW is NOT equal to $\frac{\lambda}{2}$ or λ

4

The correct answer is **B** because:

- The diffraction grating equation is given by
 - $d \sin \theta = n\lambda$
- For $n = 1$
 - $\sin \beta = \frac{\lambda}{d}$
 - So, $\beta = \sin^{-1}\left(\frac{\lambda}{d}\right)$
- From the diagram, $2\beta = \alpha$
- Therefore, $2\beta = \alpha = 2 \sin^{-1}\left(\frac{\lambda}{d}\right)$

5

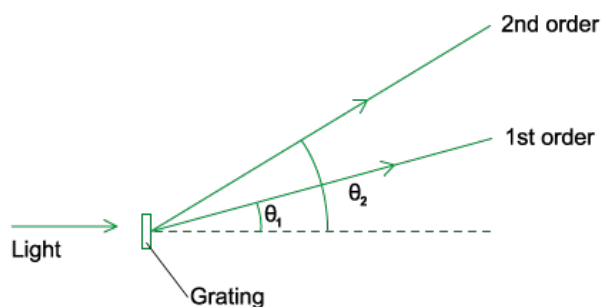
The correct answer is **D** because:

- The fringe separation equation is given by
 - $x = \frac{\lambda D}{a}$
- Where:
 - x_1 is the fringe separation = 4.0 mm, which changes to x_2
 - λ_1 is the wavelength = 600 nm, which changes to $\lambda_2 = 400$ nm
 - a is the slit separation, which changes to $2a$
 - D is the distance between the double slit and the screen, which does not change
- So, since distance D does not change, we should rearrange for D :
 - $D = \frac{xa}{\lambda}$
 - $D = \frac{x_1 a_1}{\lambda_1} = \frac{x_2 a_2}{\lambda_1}$
- Now, we need to rearrange for the new fringe spacing x_2 :
 - $x_2 = \frac{x_1 a_1 \lambda_2}{\lambda_1 a_2} = \frac{(4.0)(400)a}{600(2a)} = 1.33 \text{ mm}$
- Therefore, the new fringe spacing is 1.3 mm

6

The correct answer is **B** because:

- The diffraction grating equation is given by
 - $d \sin \theta = n\lambda$
- When $n = 1$
 - $\sin \theta_1 = \frac{\lambda}{d}$
 - The angle $\theta_1 = 15.4^\circ$, so, $\sin 15.4 = 0.2656$
- When $n = 2$
 - The angle is θ_2
 - $\sin \theta_2 = \frac{2\lambda}{d}$
- $\frac{\lambda}{d}$ is the same at both the first-order and second-order maxima, so:
 - $\frac{\lambda}{d} = \sin \theta_1 = \frac{1}{2} \sin \theta_2 = 0.2656$
 - $\sin \theta_2 = 2 \times 0.2656 = 0.5311$
 - $\theta_2 = \sin^{-1} 0.5311 = 32.08^\circ$
- Therefore, the angle between the first order and second order maxima is:
 - $\theta_2 - \theta_1 = 32.1^\circ - 15.4^\circ = 16.7^\circ$

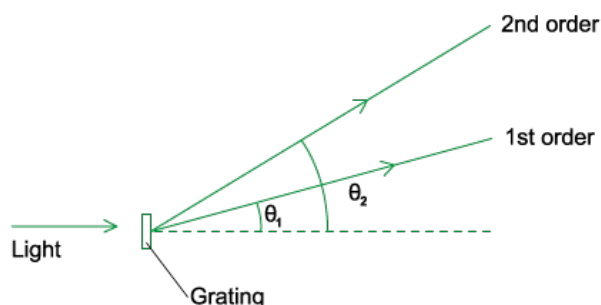


7

The correct answer is **A** because:

- The fringe separation equation is given by

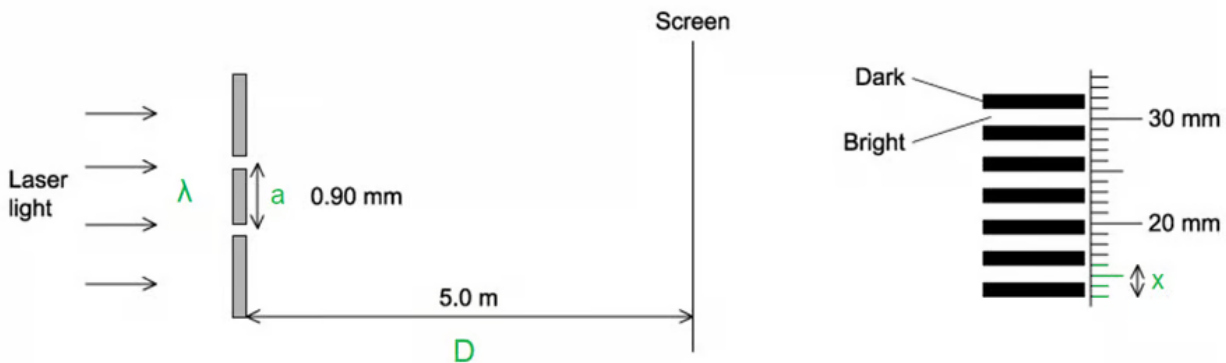
- $\sin \theta_2 = \frac{2\lambda}{d}$
- $\frac{\lambda}{d}$ is the same at both the first-order and second-order maxima, so:
 - $\frac{\lambda}{d} = \sin \theta_1 = \frac{1}{2} \sin \theta_2 = 0.2656$
 - $\sin \theta_2 = 2 \times 0.2656 = 0.5311$
 - $\theta_2 = \sin^{-1} 0.5311 = 32.08^\circ$
- Therefore, the angle between the first order and second order maxima is:
 - $\theta_2 - \theta_1 = 32.1^\circ - 15.4^\circ = 16.7^\circ$



7

The correct answer is **A** because:

- The fringe separation equation is given by
 - $x = \frac{\lambda D}{a}$
- Where:
 - x is the fringe separation = 3.0 mm
 - a is the slit separation = 0.90 mm
 - D is the distance between the double slit and the screen = 5.0 m
 - λ is the wavelength of the light
- Rearranging for wavelength λ
 - $\lambda = \frac{xa}{D}$
 - $\lambda = \frac{(3.0 \times 10^{-3})(0.90 \times 10^{-3})}{5.0} = 5.4 \times 10^{-7} \text{ m}$
- Therefore, the wavelength of the light is equal to 540 nm



8

The correct answer is **B** because:

- The diffraction grating equation is given by
 - $d \sin \theta = n\lambda$
- Since the first order maxima was seen at 30°
 - $d \sin 30^\circ = \lambda$
 - $\lambda = \frac{d}{2}$
- The fringe separation equation is given by
 - $x = \frac{\lambda D}{a}$
- Where:
 - x is the fringe separation
 - a is the slit separation = $500d$
 - D is the distance between the double slit and the screen = 1.0 m
 - λ is the wavelength of the light = $\frac{d}{2}$

the wavelength of the light $\frac{2}{2}$

- So, the fringe separation is equal to
 - $x = \frac{\lambda D}{a} = \frac{d}{2(500d)} = \frac{1}{1000} = 1 \times 10^{-3} \text{ m}$
- Therefore, the fringe separation is 1 mm

9

The correct answer is **A** because:

- The diffraction grating equation is given by
 - $d \sin \theta = n\lambda$
- Where:
 - $n = 1$, since these are first-order lines
 - d is the split spacing $= 1.15 \times 10^{-6} \text{ m}$
 - θ is the angle between the light and the horizontal $= \frac{60^\circ}{2} = 30^\circ$
- Rearranging for wavelength λ
 - $\lambda = \frac{d \sin \theta}{n} = 1.15 \times 10^{-6} (\sin 30) = 5.75 \times 10^{-7} \text{ m}$
- Therefore, the wavelength of the light is 575 nm

10

The correct answer is **C** because:

- The diffraction grating equation is given by
 - $d \sin \theta_n = n\lambda_n$
- When the second-order and third-order spectra overlap:
 - $\theta_2 = \theta_3$
 - $d \sin \theta_2 = 2\lambda_2 = d \sin \theta_3 = 3\lambda_3$
 - Therefore, $2\lambda_2 = 3\lambda_3$
- Rearranging for λ_3 :
 - $\lambda_3 = \frac{2}{3}\lambda_2 = \frac{2}{3} \times 600 = 400 \text{ nm}$
- So, a wavelength of 400 nm would appear at the same angle as the 600 nm wavelength

